BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Selection: IF Statements

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. When is A AND B true?

2. What does CPU cache hold?
   A) Only saved personal files
   B) Only website passwords
   C) Often-used data and instructions
   D) Only the next address

3. "A register is a large external drive used for backups." Correct this. Explain why.

4. The next instruction is at address 240. An addition gives 18. What do the accumulator and Program Counter hold? Explain the difference.

Today's objectives

Use IF statements to make decisions. Use selection and if in a practical scenario.

Key vocabulary - Read, cover, recall

Selection

Selection lets a program follow different instructions according to a condition.

IF

An IF block executes its instructions only when the condition is true.

Comparison

The Boolean test must compare values rather than assign a new value.

ENDIF

ENDIF marks the end of the IF statement so later instructions can run normally.

BLACK PEN: Write one definition in your own words.

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

What is Selection?

Selection - Selection lets a program follow different instructions according to a condition.

In this example

A theme-park ride height rule uses if; An IF block executes its instructions only when the condition is true.

Follow the effect

The Boolean test must compare values rather than assign a new value. Explain how this changes the eventual program result.

Check understanding

ENDIF is used for a specific reason: ENDIF marks the end of the IF statement so later instructions can run normally.

I Do - Worked model

A theme-park ride height rule is being written. Explain how selection and if are used to make the program work. [4 marks]

- We are designing a theme-park ride height rule. Identify which data and decisions are essential.
- Selection [1] - Selection lets a program follow different instructions according to a condition. [1].
- IF [1] - An IF block executes its instructions only when the condition is true. [1].
- Two linked points for each technique give four mark points.

We Do - Guided practice

1. In a school competition entry, explain how selection helps the program. [2 marks]

2. For a sports pass check, explain why if matters. [2 marks]

3. When designing a exam pass notification, explain how comparison and endif should be used. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the IF condition that switches on ventilation above 30 degrees.

```
temperature = 32
if ... then
print("Fan on")
endif
```

You Do - Independent exam questions

Q2. State two programming terms or techniques relevant to this lesson. [2 marks]

Q3. A cinema age check needs to work correctly. Explain how if and comparison would help. [4 marks]

Q4. A pupil is programming a library opening status. Explain the roles of selection, comparison and endif in their solution. [6 marks]

Exit Ticket - True or False?

Selection lets a program follow different instructions according to a condition.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Only when both A and B are true.

2. C - Often-used data and instructions

3. A small, fast storage location inside the CPU. It holds data or addresses in use.

4. The accumulator holds the result, 18. The PC holds the next instruction's address, 240.

I Do - Explain the model

We are designing a theme-park ride height rule. Identify which data and decisions are essential.

Selection [1] - Selection lets a program follow different instructions according to a condition. [1].

IF [1] - An IF block executes its instructions only when the condition is true. [1].

Two linked points for each technique give four mark points.

We Do - Model answers

1. Selection [1] - Selection lets a program follow different instructions according to a condition. [1].

2. IF [1] - An IF block executes its instructions only when the condition is true. [1].

3. Comparison [1] - The Boolean test must compare values rather than assign a new value. [1]. ENDIF [1] - ENDIF marks the end of the IF statement so later instructions can run normally. [1].

Code First - Answer

```
temperature = 32
if temperature > 30 then
print("Fan on")
endif
```

Expected output: Fan on

You Do - Self-mark

Q2: Selection [1]; Comparison [1].

Q3: IF [1] - An IF block executes its instructions only when the condition is true. [1]. Comparison [1] - The Boolean test must compare values rather than assign a new value. [1].

Q4: Selection [1] - Selection lets a program follow different instructions according to a condition. [1]. Comparison [1] - The Boolean test must compare values rather than assign a new value. [1]. ENDIF [1] - ENDIF marks the end of the IF statement so later instructions can run normally. [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.